

26-1-2

H.W # 7

Fall 2013

Koku Adobe

at 378.2

V_p or Benzene $P^B = 204.2 \text{ kPa}$

V_p or Toluene $P^T = 86 \text{ kPa}$

we know Raoult's Law

$$P = X_A P^B + X_B P^T$$

$$P = P_{\text{Tot}} = 101.325$$

$$101.325 = X_A 204.2 + X_B 86 \quad X_A + X_B = 1$$

$$101.325 = 204 \cdot X_A + 86 - 86 X_A \quad Y_A + Y_B = 1$$

$$X_A = 0.1296$$

$$X_A P^B = Y_A P$$

$$Y = \frac{X_A P^B}{P} = \frac{0.1296 \times 204.2}{101.325}$$

$$Y_A = 0.2612$$

(b) If $P = 101.325 = X_A P_B^{\text{sat}} + X_B P_T^{\text{sat}}$
 it will boil at $T = 358.2$
 Benzene, $P_B^{\text{sat}} = 116.9$
 Toluene, $P_T^{\text{sat}} = 46$
 $X = 0.4$

$$= 0.4 \times 116.9 + 0.6 \times 46$$

$$= 74.36 \quad \neq 101.325$$

It is will not boil.

not

$$\text{at } T = 368.2 \quad P^B = 155.7 \text{ kPa}$$

$$P_T = 63.3 \text{ kPa}$$

$$p_{\text{obtained}} = 0.4 \times 155.7 + 0.6 \times 63.3$$

$$= 100.26$$

$$\approx 101.325$$

y coming out

$$y = \frac{P^B x_B}{P}$$

$$= \frac{155.7 \times 0.4}{100.26}$$

$$y_A = 0.62 \rightarrow \text{Benzene}$$

$$\text{Toluene } y_T = 1 - y_A = 0.38$$

$$\boxed{26-2-1}$$

$$n - \text{pentane} = 40 \text{ mol } \%$$

$$P_{\text{atm}} = 101.32 \text{ kPa}$$

$$n - \text{heptane} = 60 \%$$

$$\text{mixture} = 100 \text{ mol}$$

$$V = 40 \text{ mol} \begin{cases} y_A = \% \text{ mole of } n\text{-pentane in vapor} \\ y_B = \% \text{ mole of } n\text{-heptane in vapor} \end{cases}$$

$$L = 60 \text{ mol} \begin{cases} x_A = \% \text{ mole of } n\text{-pentane in liq} \\ x_B = \% \text{ mole of } n\text{-heptane in liq} \end{cases}$$

$$x_A + x_B = 1$$

$$y_A + y_B = 1$$

mole balances

$$n x_{A1} = V y_A + L x_A$$

$$100 \times 0.6 = 40 y_B + 60 x_B$$

$$100 \times 0.4 = 40 y_A + 60 x_A$$

$$60 = 40 y_B + 60 x_B$$

$$40 = 40 y_A + 60 x_A$$

$$y_B = \frac{60}{40} - \frac{60}{60} x_B$$

$$y_A = \frac{40}{40} - \frac{60}{40} x_A$$

$$y_B = 1.5 - x_B$$

$$y_A = 1 - 1.5 x_A$$

x_A	1	0.867	0.594	0.398	0.25	0.145	0.059
y_A	1	0.984	0.925	0.836	0.701	0.521	0.271

$$y_A = 0.1063 + 2.499 x_A - 1.6595 x_A^2$$

$$1 - 1.5 x_A = 0.1063 + 2.499 x_A - 1.6595 x_A^2$$

$$1.6595 x_A^2 - 3.999 x_A + 0.8937 = 0$$

$$x_A = 3.999 \mp \sqrt{[3.999]^2 - 4[1.6595][0.8937]}$$

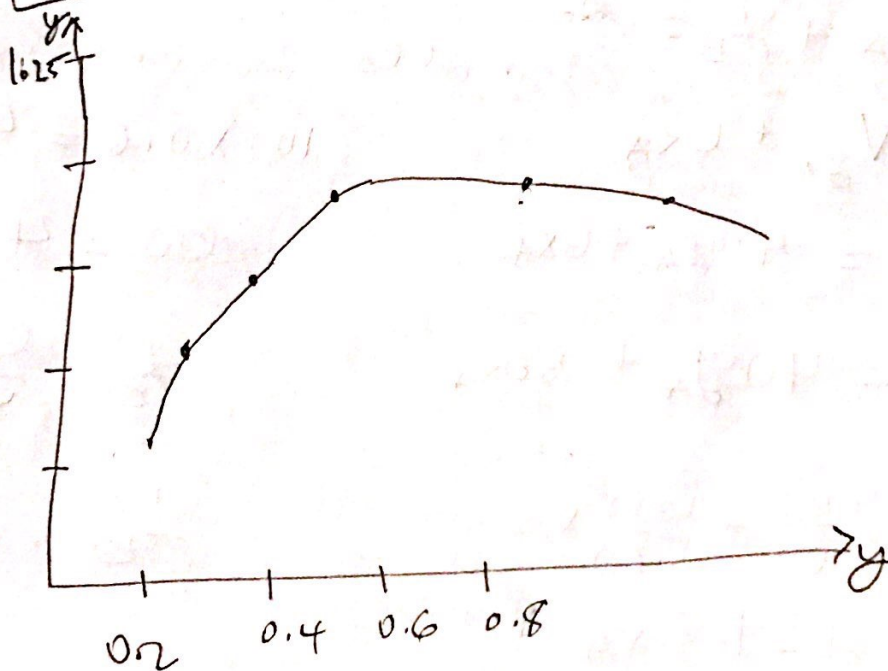
$$x_A = 0.249 \quad \checkmark$$

$$y_A = 0.626 \quad \checkmark$$

$$2 \times 1.695$$

$$x_B = 1 - x_A = 0.751$$

$$y_B = 1 - y_A = 0.374$$



$$\begin{bmatrix} 26 & -3 & -2 \end{bmatrix}$$

(a) $L_1 = 100 \text{ kg mol}$ $L_2 = 60 \text{ kg mol}$

$X_1 = 0.6$

$x_2?$

$X = L_1 - L_2 = 40$

$\ln \frac{L_1}{L_2} = \ln \frac{100}{60} = 0.5108 = \int_{x_2}^{x_1} \frac{dx}{y-x}$
 interpolating the Area under the curve
 $= 0.518$

$x_2 = 0.405$

$L_1 X_1 = L_2 X_2 + (L_1 - L_2) y_{ave}$

$100 \times 0.6 = 60 \times 0.405 + (100 - 60) y_{ave}$

$y_{ave} = 0.892$

(b) Eq on flash distillation

$F = 100$ $y = 40$ $F - y = 60$

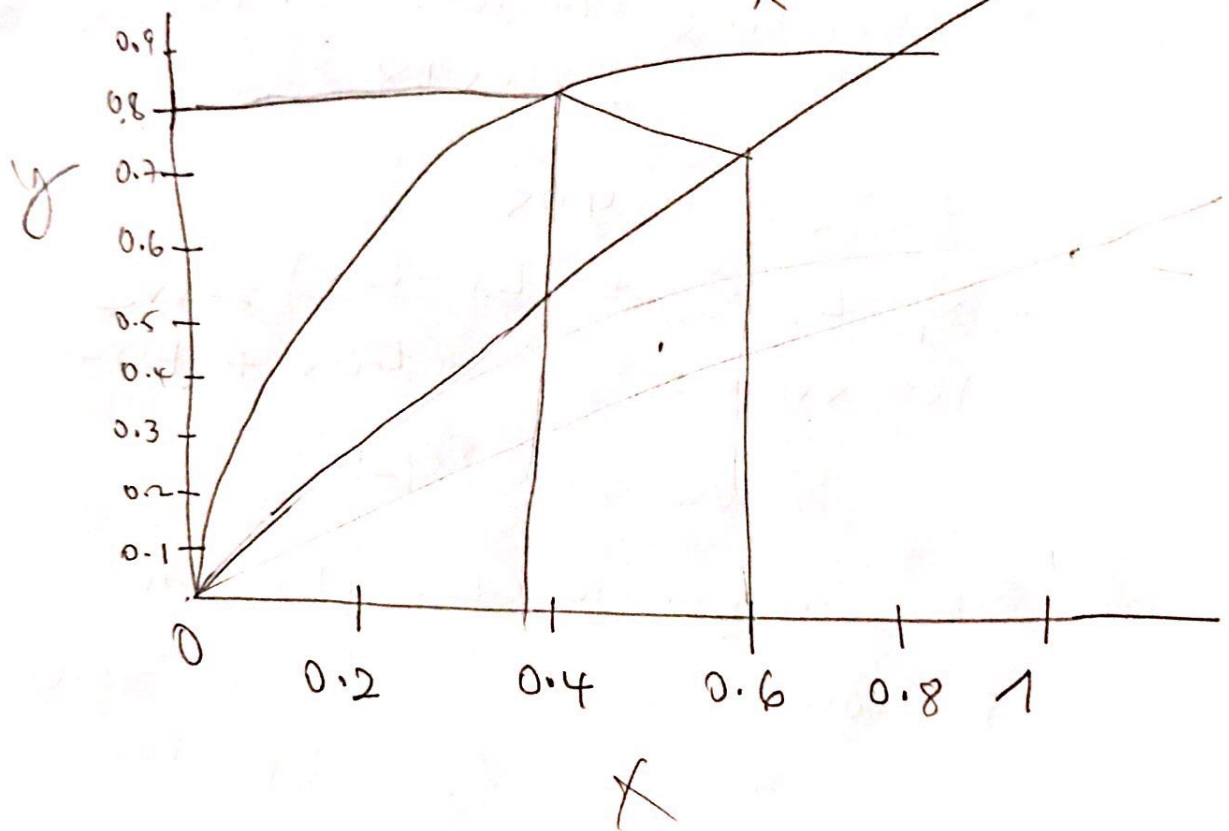
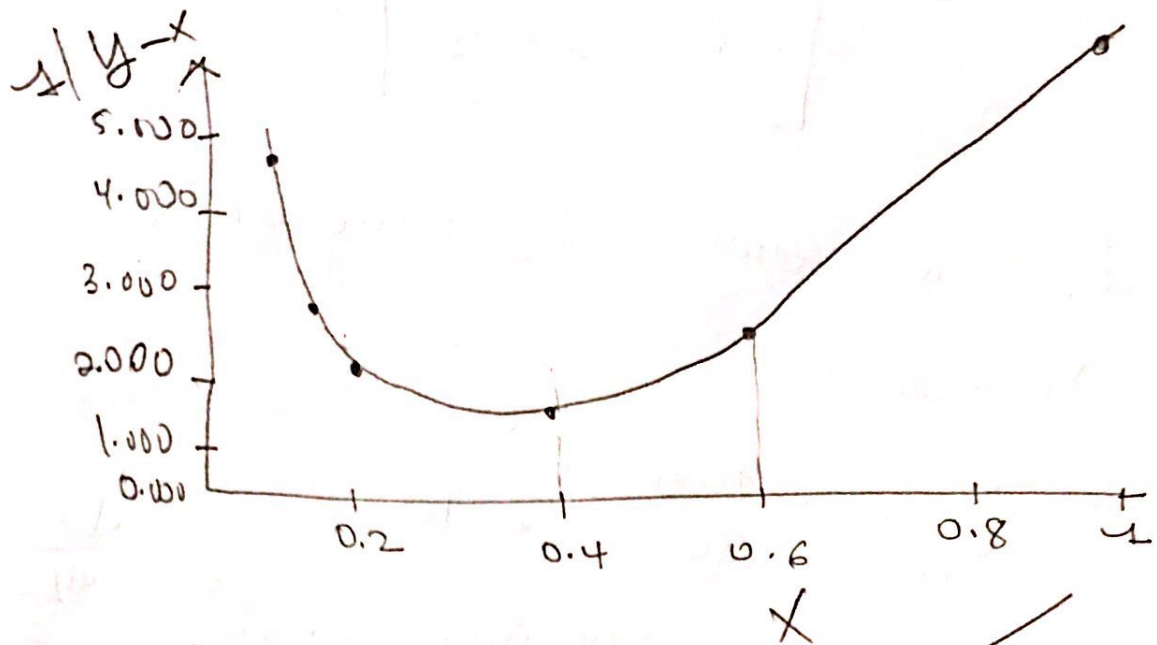
$F x_1 = V y + (F - V) x$

$100 \times 0.6 = 40 y + 60 x$

$60 = 40 y + 60 x$

Trial & error solution

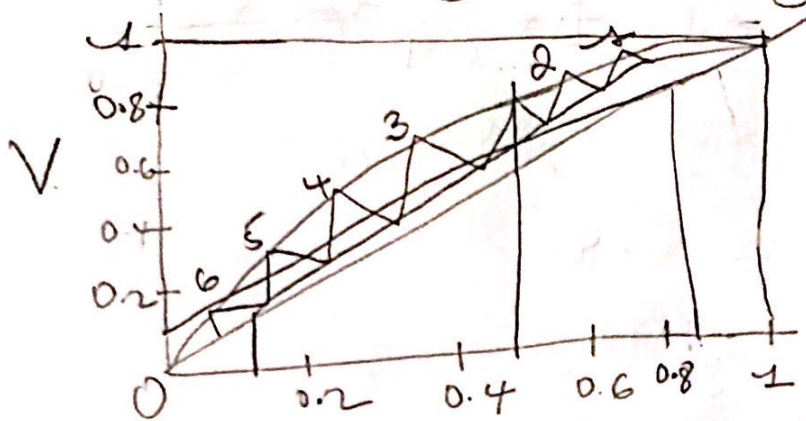
$y = 0.5$, $60 = 40 \times 0.5 + 60 x \Rightarrow x = 0.667$
 $y = 0.8$, $60 = 40 \times 0.8 + 60 x \Rightarrow x = 0.467$
 $y = 0.9$, $60 = 40 \times 0.9 + 60 x \Rightarrow x = 0.4$
 $y_2 = 0.854$ $x_2 = 0.430$



26-4-2

Solution

Benz-Tolu System



$$V = \frac{R}{1}x + \frac{x_D}{1}$$

$$V - \text{Ent} = \frac{x_D}{1} = \frac{1}{1}$$

$N = 5.9$ then trays + reboiler = $n+1$

$n = 4.9$ theo. trays

• Total balance

$$F = D + B$$

• more volatile component balance

$$X_F F = X_D D + X_B B$$

$$0.5(100) = 0.9 D + 0.1 B$$

• Solving

$$D = 50 \text{ kmol s/h}$$

$$B = 50 \text{ kmol/h}$$

$$n_{\min} = 3.9$$

$$R_m = 0.82$$

$$26.4 - 2$$

$$D = 85.2972 \text{ mol/h} \quad F = D + B$$

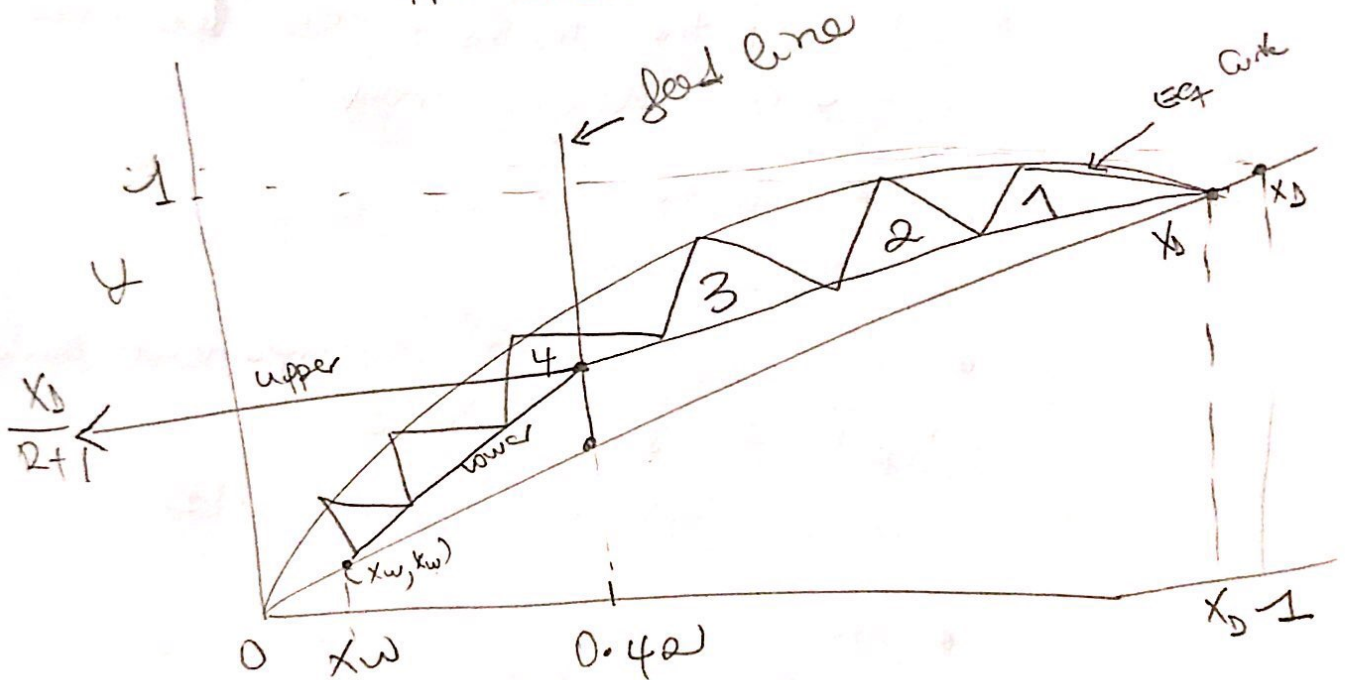
$$B = 114.7028 \text{ mol/h} \quad F \cdot z_f = D \cdot y_H + B \cdot x_B$$

$$F = 200 \text{ mol/h}$$

$$z_f = 0.42$$

$$y_H = 0.97$$

$$x_H = 0.011$$



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② $F = 200 \text{ mol/hr}$

Compos- $= X_{HF} = 0.42$

$X_{BF} = 0.58$

$X_{HD} = 0.97$

Bottom production

$X_{HB} = 0.011$

$F = D + B$

$D + B = 200 \leftarrow \textcircled{1}$

$X_{HD} \cdot D + X_{HB} \cdot B = X_{HF} \cdot F$

$0.97D + 0.011B = 0.42 \times 200$

$0.97D + 0.011B = 84 \leftarrow \textcircled{2}$

$D = 85.297 \leftarrow \textcircled{1} \text{ and } \textcircled{2}$

Distillate

$D = 114.70 \text{ mol/hr}$

Bottom product

⑥ Feed = 40% Liq & 60% vapor

$q = 0.4$

$y = \frac{q}{q-1} x - \frac{X_F}{(q-1)}$

$y = \left(\frac{0.4}{0.4-1} \right) x - \frac{0.48}{(0.4-1)}$

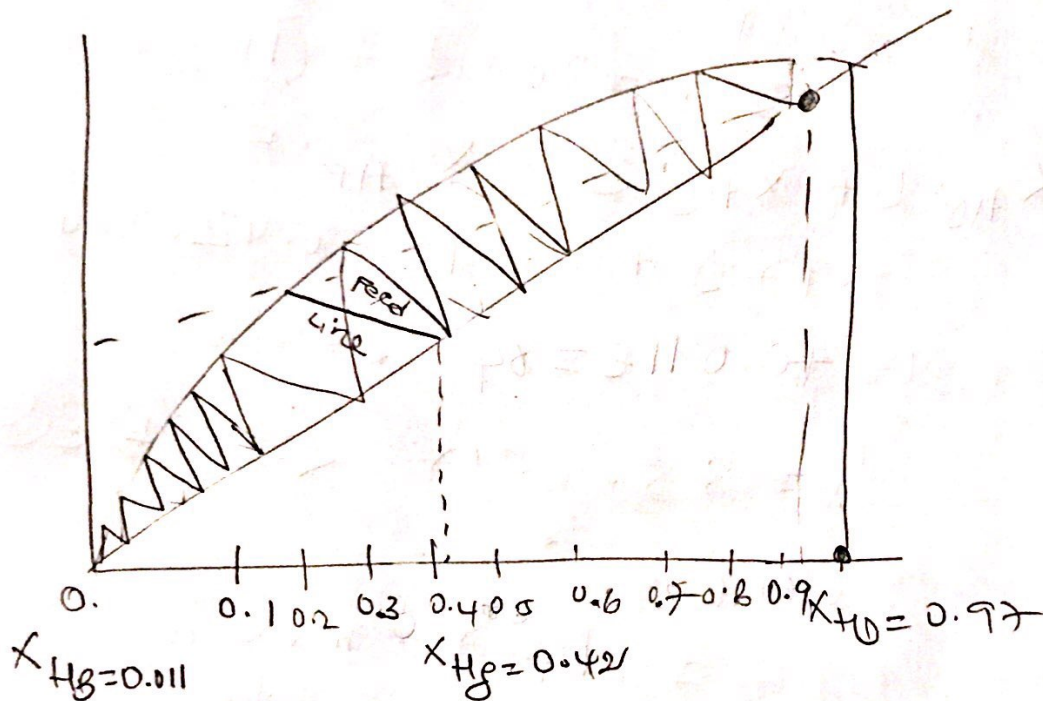
$y = -0.666x + 0.8$

$$y = \left(\frac{R}{R+1} \right) x + \frac{x_D}{R+1}$$

$$\frac{x_D}{R_m + 1} = 0.36$$

$$\frac{0.97}{0.36} = R_m + 1$$

$$R_m = 1.694$$



(c) Minimum
or trays = $8 - 1 = 7$

(d) $y = \left(\frac{2.5}{2.5+1} \right) x + \frac{0.97}{(2.5+1)}$

$$y = 0.714x + 0.277$$

(a) we know 26-4-9

$$F + S = V + B$$

$$x_F F = x_B B + y_D V$$

$$(0.033)(100) = x_B (100) + y_D V$$

$$y_D V = (0.99)(0.033)(100)$$

$$x_B = (100) - (0.01)(100)(0.033)$$

$$x_B = 0.00033$$

$$x_F = 0.033$$

$$y_D = 0.27$$

$$y_D V = (99)(0.033)$$

$$V_{\min} = (99)(0.033) / 0.27$$

$$V_{\min} = 12.1 \text{ mol/hr} = S_{\min}$$

Minimum 0.121 mol Steam/mol feed

(b)

$$S = 2 S_{\min} = 24.2 \text{ mol}$$

$$100(x_F - x) = 24.2(y_D - y)$$

$$y = 0$$

$$x_B = 0.00033$$

$$y_D = (100)(0.033 - 0.00033) / 24.2 = 0.135$$

$$100(0.033 - x) = 24.2(0.135 - y)$$

$$\# \text{ equilibrium stages} = 4.7378$$

